PART I

Resolution and free dynamics

Exact packet algebra becomes field dynamics only after reconstruction, completion, and domain control are supplied. The argument moves from finite transfers to a torus Hamiltonian and then identifies its full unitary group.

IN THIS PART

Projectors, Packets, and Transfer 2

Fourier Packets and the Continuum Dirac Operator 35

The Dirac Group and Its Generator 67

PART II

Gauge transport and gravity

A compatible link law is not yet a coupled field evolution. Gauge transport introduces ordered paths and curvature. Gravity adds dynamical geometry, constraints, boundary conditions, and a distinct continuum comparison.

IN THIS PART

Gauge Covariance and Geometric Refinement 75

Constrained Gravity and Strong Fields 107

PART III

Arithmetic and polyphase refinement

Arithmetic registers provide another exact resolution of a quantum state. Compatibility determines when refinement preserves the dynamics. Eliminating hidden channels produces self-energy and memory, while infinite depth requires its own analytic domain.

IN THIS PART

Adic and Polyphase Quantum Refinement 161

PART IV

Matter and cosmological transport

A mathematical refinement fiber becomes a physical proposal only after a coupling and production history are specified. The energy-budget obstruction is examined before the distribution is propagated through cosmological observables.

IN THIS PART

Covariant Refinement-Fiber Dynamics 213

Cosmology, Dark Sectors, and Structure Formation 245

PART V

Visual studies

The figures follow the same objects through explicit examples. Their purpose is to make algebra, geometry, approximation, and failure modes inspectable. Every numerical illustration retains its stated input and does not replace the corresponding theorem.

IN THIS PART

Foundations Visual Atlas 396

Dirac Continuum Visual Atlas 411

Gauge Visual Atlas 426

Gravity Visual Atlas 441

Adic and Polyphase Visual Atlas 457

PART VI

Verification and computational evidence

A proof, a compiler certificate, a numerical check, and a physical interpretation answer different questions. The following appendices identify their exact scope. Source and toolchain identifiers appear here only where they are needed to inspect or reproduce the evidence.

IN THIS PART

Verification Scope and Source Conventions 474

Foundations Formalization Companion 477

Dirac Formalization Companion 491

Gauge Formalization Companion 508

Gravity Verification Companion 522

Adic Verification Companion 540

Fiber Model: Computational Evidence 557

Cosmology: Reproducibility and Data 564